

The neural correlates of domain-general visual ability

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Abstract

People vary in their general ability to compare, identify, and remember objects. Research using latent variable modeling identifies a domain-general visual recognition ability (called o) that reflects correlations among different visual tasks and categories. We measure associations between a psychometrically-sensitive measure of o and a neurometrically-sensitive measure of visual sensitivity to shape. We report evidence for distributed neural correlates of o using functional and anatomical regions-of-interest (ROIs) as well as whole brain analyses. Neural selectivity to shape is associated with o in several regions of the ventral pathway, as well as additional foci in parietal and premotor cortex. Multivariate analyses suggest the distributed effects in ventral cortex reflect a common mechanism. The network of brain areas where neural selectivity predicts o is similar to that evoked by the most informative features for object recognition in prior work, showing convergence of two different approaches on identifying areas that support the best object recognition performance. Because o predicts performance across many visual tasks for both novel and familiar objects, we propose that o could predict the magnitude of neural changes in task-relevant areas following experience with specific task and object category.

Introduction

Many human endeavors require visual object recognition skills. Some people excel at specific visual tasks, such as fingerprint matching or bird recognition, because of relevant experience. But people also vary in their general ability to compare, identify, and remember objects. Just like Spearman's g (Spearman 1927) reflects correlations among a variety of cognitive tests, a recently identified factor called o reflects correlations among visual object recognition tests (Richler et al. 2019; Grown et al. 2022). Individuals with a high o excel at both perceptual and memory tasks, with both artificial and familiar object categories. This domain-general ability is only weakly correlated with g (Richler et al. 2019). Individual differences in o can help account for performance in a range of occupations (Sunday et al. 2018; Holmes et al. 2020; Chang and Gauthier 2021). Here, we investigate for the first time the functional neural correlates of o .

In searching for the brain areas where neural responses to objects can predict o , we approached our fMRI task design using the same principles used to create valid and reliable behavioral measurements. Because the behavioral ability we target is domain-general, we opted for a design in which we could measure neural sensitivity to shape in a broad manner (this refers to the construct coverage of the measure, which is part of construct validity). In fMRI studies measuring the neural correlates of behavioral performance for a specific category, many visually homogenous stimuli from one category are typically used (e.g., birds or cars, or novel objects from one category), and are compared to objects from one or more control categories (Gauthier, Skudlarski, et al. 2000; Bilalić et al. 2011; McGugin et al. 2012). Another approach, pattern analysis, is a powerful way to measure the brain's sensitivity to the differences between similar objects. Yet, robust decoding requires many trials with objects of the same category (VanRullen and Reddy 2019). Here, we instead wanted to measure neural sensitivity to visual differences for objects across a wide range of distinct shapes, to capture a general neural sensitivity to object shape that could support o .

To meet our goals, we opted to use fMRI-adaptation (Grill-Spector et al. 1999a; Gauthier, Tarr, et al. 2000) to measure neural sensitivity to differences in the shape of a range of different novel objects, then relate this brain response to o . We chose to use novel objects to avoid any variability due to experience (Richler et al. 2019). fMRI-adaptation can reveal sensitivity in

category-selective areas to varying properties of the images (e.g., size, viewpoint, identity, etc.). For instance, face-selective and letter-selective areas show less repetition suppression with changes in face identity or letter font, but only for their preferred category (Gauthier, Tarr, et al. 2000). While fMRI-adaptation typically is done with objects of the same category (Natu et al. 2016), we adapted the task and averaged the adaptation measured for a variety of novel objects from different categories. Thus, in the same way our behavioral index of o combines measurements across several tasks and categories, this neural index of visual sensitivity combines measurements of sensitivity across categories. For these neural measurements, a task was chosen to be orthogonal to the adaptation manipulation, to assess how task-irrelevant neural sensitivity to shape is associated with behavioral ability.

In addition to these considerations about the validity of our neural measure, we also wanted to optimize it for reliability. Reliable measurements for a variable is necessary before exploring its relation to other measures (Ross et al. 2015; Hedge et al. 2018). Just as the reliability of behavioral abilities is increased by using a range of trial difficulty along the variable of interest (Richler et al. 2014), we sought to increase the sensitivity of our neural measure by varying the similarity of pairs of shapes included in the task. In this way, differential adaptation for the shapes that are more different can inform on variability between voxels at the low range of sensitivity to shape, while differential adaptation for shapes that are more similar can inform on variability between voxels at the high range of sensitivity to shape. In sum, we characterize associations between a psychometrically-sensitive measure of domain-general object recognition ability and a neurometrically-sensitive measure of sensitivity to shape.

No work to date has addressed the functional neural substrates of o . However, there is research on the functional responses associated with some of the specific indicator tasks that contribute to o . Specifically, the novel objects we use here in the scanner and the subordinate-level tasks underlying our behavioral measurements are similar to those used in many expertise studies (Gauthier et al. 1999; Wong et al. 2009, 2012). When selectivity for an object category is measured in the scanner and related to the behavioral performance for objects from the same category, a correlation is obtained in a number of visual areas, with the most robust expertise effects found in the

right middle fusiform face area, or FFA2 (McGugin et al. 2014, 2018; Golarai et al. 2017; Ross et al. 2018). In addition, behavioral performance in face discrimination relates to neural sensitivity for face identity measured with fMRI-adaptation in face-selective regions and not in object selective regions (Natu et al. 2016). As such, functional neural correlates of domain-specific recognition ability, especially for tasks that involve individuation of objects within a category, converge on face-selective regions for both face and non-face objects (Burns et al. 2019). Face-selective regions could therefore be a plausible locus for a correlation between neural responses to novel objects and behavioral performance with these novel objects. One argument against *o* specifically tapping into face-selective areas is that when expertise studies use familiar objects (like cars or birds), individual variability may be largely driven by experience. Therefore, in such studies, domain-general effects are avoided because both neural and behavioral measures use other objects as controls. However, it is important to note that in studies in which subjects are trained to identify novel artificial objects, all subjects are provided the same amount of training and the ensuing variability in performance is nonetheless correlated with responses in face-selective areas (Gauthier et al. 1999; Wong et al. 2009, 2012). Therefore, it is possible that variability in responses to objects in face-selective areas expresses experience-independent differences in ability in addition to being sensitive to experience.

Other work predicts a more distributed neural substrate for *o*. Training on visual tasks can result in widespread changes in occipitotemporal cortex (Kourtzi et al. 2005; Sigman et al. 2005). Experience with the same objects can result in different patterns of changes depending on the type of training provided (Wong et al. 2009, 2012). If training can result in very different patterns of neural recruitment depending on experience, properties of neuronal responses that predict the potential to learn may be distributed across several visual areas. Likewise, neural object representations depend on the task performed, more so in ventral category-selective areas than in LO or early visual cortex (Harel et al. 2014). A domain-general ability like *o*, which by definition applies across tasks, may flexibly tap into neural responses in all visual areas that can support object recognition. Finally, one line of work that may best foreshadow the neural pattern of activity associated with *o* is research on the neural responses evoked by visual features known to best support object recognition. Visual fragments selected to be maximally informative for object recognition, when compared to similar but

less informative fragments (Ullman et al. 2002, 2016) engage a large set of object and face-selective occipitotemporal areas, as well as parietal and premotor cortex (Lerner et al. 2008; Holzinger et al. 2019). The selectivity for informative visual features of specific categories was not limited to the most relevant category-selective areas, suggesting they reflect a heterogeneous neural population that could support a domain-general ability. Therefore, individual differences in o may have neural correlates that are similar to those obtained for the features that best support object recognition.

Materials and Method

Participants. Participants were recruited from a previous behavioral study (Sunday et al. 2021) to sample a range of object recognition abilities (see Supplementary Figure S1). Thirty-seven participants were scanned roughly 1-9 months after completing the behavioral study, though scanner malfunction led to functional data collection for only thirty-six subjects and one subject was omitted due to excessive head motion ($N=35$, 19 females, mean age = 21, $SD = 3.8$). Although we use Bayesian statistics, we estimated the sample size using a standard power analysis with G*Power software v3.1 (Faul et al. 2009) based on an effect size in (Natu et al. 2016), showing a correlation between neural sensitivity and face perceptual discriminability in inferior occipital gyrus of $r = 0.47$). A sample of 33 participants is required to detect such an effect at an alpha of .05, with 90% power. All participants provided informed consent according to the guidelines of the institutional review board at Vanderbilt University.

Behavioral estimates of o . The ability o was measured as a higher order factor that could account for 88% of the variance among 12 distinct tests. These included a matching task and a learning task (see Figure 1a) with each of 3 categories of artificial objects (Greebles, Ziggerins and Sheinbugs, Richler et al. 2017) and each of 3 categories of familiar objects (passerine birds, planes and Transformer toys). Behavioral estimates of o for our 35 participants were factor scores (calculated with the maximum a posteriori method (Bock and Aitkin 1981) from a confirmatory factor analysis model using a larger number of participants ($N=284$) to estimate all parameters (Sunday et al. 2021). In this work, o was essentially identical ($r=.98$) whether measured with familiar or novel objects.

Factor scores for the higher order factor o (Figure 1b) in our participants had a mean of 0, a SD of .29, a range = [-0.80-.43], skewness of -0.58 and kurtosis of -.05. The distribution of factor scores for our participants and the larger sample is provided in the Supplementary materials.

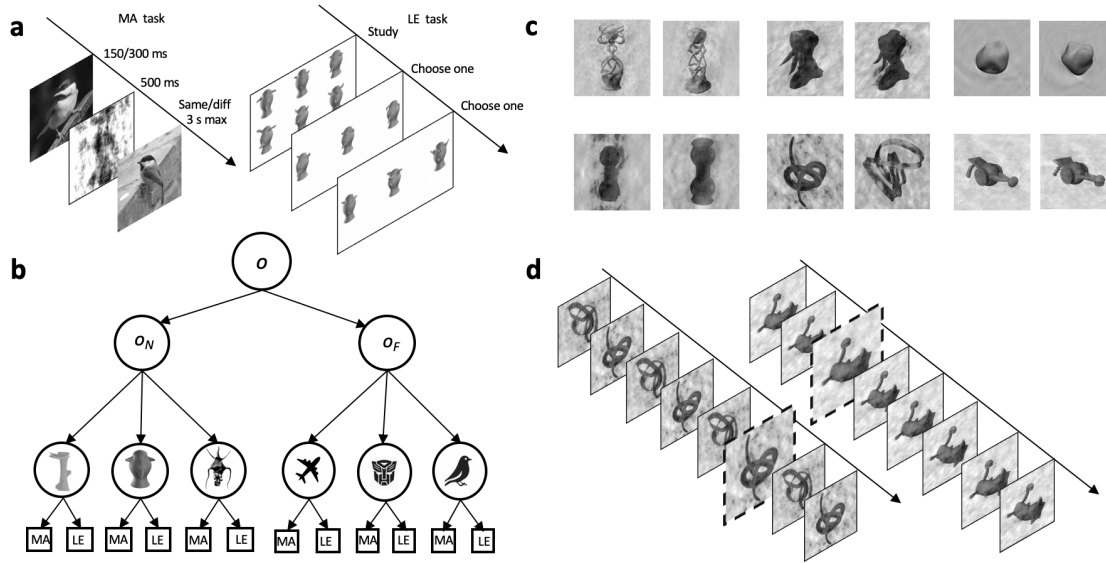


Fig. 1 | Tasks outside and inside the scanner. (a) Sample trials for the Matching task (MA) and Learning task (LE) from the behavioral battery used to estimate o . On each trial of the MA task, participants decide if a pair of sequentially presented images depicts the same object (here the same bird species), regardless of image differences. In the LE task, participants study 6 target objects, and on each trial chose which of 3 images depicts one of the 6 target objects, regardless of image differences. **(b)** Schematic of the structural equation model used to estimate o , on the basis of performance for MA and LE tasks for 3 categories of novel objects (Ziggerins, Greebles and Sheinbugs) and 3 categories of familiar objects (planes, transformer toys and passerine birds). Factor scores for o were estimated based on data from 284 participants (Sunday et al., 2021). We use the third order factor scores for o , but factor scores for novel and familiar objects were almost perfectly correlated ($r=.98$). **(c)** Examples of pairs of novel objects used in the fMRI-A task. Top row, left to right: 3D printed abstract chess pieces, YUFOs from http://wiki.cnbc.cmu.edu/Novel_Objects, blobs used with permission from Erez Freud. Bottom row, left to right: dog toys, abstract sculpture, Fribbles from http://wiki.cnbc.cmu.edu/Novel_Objects. **(d)** Sample blocks for the alternating (left) and repeating (right) conditions used during fMRI. Each stimulus was presented for 900 ms with a 100 ms inter-stimulus interval. Dashed outlined images represent a change in size that occurred once per block and was relevant for the orthogonal task. The size change is exaggerated for illustrative purposes. All low-level features were equated within each category of object using the SHINE algorithm. In the repeating condition, the same object was alternated in two orientations (15 deg in the picture plane). In the alternating condition, two different objects from the same category were presented in alternation. The objects and backgrounds changed in both conditions for each image presentation.

Scanning parameters. Participants were scanned on a Philips 7-Tesla (7T) Achieva human magnetic resonance scanner with a 32-channel parallel receive array coil (Nova). A 3D TFE (Turbo Field Echo) acquisition sequence with sensitivity encoding (TR = 4.3 ms, TE = 1.90 ms (minimum), flip angle = 7°, sagittal plane acquisition, for 7 subjects: FOV = 256 mm, matrix size = 256 x 256, 170 slices, slice gap = 0 mm, for 30 subjects: FOV = 256 mm², matrix size = 256 x 256, 200 slices, slice gap = 0 mm, for an isometric voxel size of 1 mm³) was used to acquire high-resolution T1-weighted anatomical volumes. All functional scans were acquired using standard gradient-echo echoplanar T2*-weighted imaging (TR = 2000 ms, TE = 25 ms, flip angle = 65°, axial plane acquisition, FOV = 240 mm², matrix size = 80 x 80, slice gap = 0 mm, for an isometric 3 mm³ voxel). Following 5 dummy scans, for 31 participants, ascending interleaved slices were acquired (31 participants: 40 slices; 1 participant: 42 slices, 4 participants: 45 slices). Participants were scanned in the following order: anatomical, functional localizer, adaptation runs (4 runs). Other scans with diffusion and resting-state were acquired but are not reported here. The total scan time was approximately 1 hour and 15 minutes. For cases in which the participant appeared to move or fall asleep (6 runs) or a scanner error occurred (2 runs), the run was redone within the same session.

Functional localizer. Participants completed one functional MRI run to localize face-, object-, and body part-selective regions. Our goal was to identify the functional peak of regions-of-interest (ROIs), as we have shown that individual differences in neural selectivity are most reliable at the peak of selectivity (McGugin and Gauthier 2016). That work also showed highly reliable responses in these peaks with a single localizer scan. Participants were presented with 21 16-second blocks of faces, body parts (with no faces), household objects, and scrambled images using Matlab (Higham and Higham 2016) and Psychtoolbox-3 software (Brainard and Vision 1997). Each block consisted of 16 trials in which a stimulus is presented for 900 ms followed by a 100 ms fixation. Stimuli included greyscale images of 35 faces, 35 objects, 37 scrambled images and 54 body parts shown on a white background. A specific image repeated 1-2 times per block and participants indicated with a button press when they saw a repeat. Stimuli appeared randomly in three different sizes, and the same size never repeated in immediate succession. The run lasted 5.6 minutes.

Functional adaptation stimuli. Stimuli used in the adaptation runs represented a wide range of novel object categories to best estimate domain-general object recognition ability. An initial set of 87 novel objects collected from several sources was pared down to 52 object pairs from 26 novel object categories (2 objects per category) based on pilot behavioral data (see Supplementary material). The reliability of the hits – false alarms scores on matching trials gathered for this subset of pairs was high (Cronbach alpha = 0.91). This suggests these object pairs as a set, ranging from very distinct to very similar, can tap into a sufficient range of visual ability in behavioral measurements and therefore we hoped could provide similar sensitivity for neural measurements. All images were transformed to greyscale and equated for low-level images properties using the Matlab SHINE toolbox (Willenbockel et al. 2010). The algorithm was used to equate each pair of alternating -or identical but rotated- images in their Fourier amplitude and luminance histograms through 5 iterative repetitions (this was sufficient as the image pairs were highly similar to start with). Two versions were thus created for each object to be repeated, with each object rotated 15 deg. in the picture plane and also differing in the SHINED background.

Functional adaptation stimulus presentation and task. In the scanner, each category of novel objects was presented once in a repeating block and once in an alternating block (Figure 1C and 1D). To minimize adaptation across the entire scan (e.g. to avoid showing a repeating and an alternating block from the same domain in immediate succession), all 26 object pairs were shown once (domains 1 to 26 across runs 1 and 2) and then shown again in the same order (1 to 26 again across runs 3 and 4). In this way, 25 object pairs were shown between each presentation of one domain, minimizing any domain-specific adaptation effects (Grill-Spector and Malach 2001). Repeating and alternating blocks were interleaved. Each block included the presentation of eight images. On repeating blocks, the same object was shown on each trial (though the image did increase in size once per block, see below). On alternating blocks, an object pair was shown in an ABABABAB format (Figure 1D). For repeating blocks, every other image was rotated 15-degrees clockwise so that while the same exact object was shown on subsequent trials, the same exact image (in the same

orientation) was not (Figure 1D). Each 1000 ms trial consisted of a 900 ms stimuli presentation followed by a 100 ms inter-stimulus interval. Fourteen fixation blocks of 6000 ms were randomly inserted in each run. During these blocks, a red fixation cross was presented in the center of the screen on a grey background. The order of conditions in the adaptation task was held constant across participants.

Participants performed a size-detection task in which they pressed a response button when the stimulus size increased (425 pixel x 425 pixel to 500 pixel x 500 pixel for 7 subjects, 525 pixel x 525 pixel to 600 pixel x 600 pixel for the remaining subjects for which stimuli were presented using a different computer). The FFA has shown invariance to size changes of up to three-fold increases (Andrews and Ewbank 2004), and the LOC shows invariance to much larger size changes (e.g., from $7^\circ \times 7^\circ$ to $22^\circ \times 22^\circ$ (Grill-Spector et al. 1999b). Thus, this size change should not be large enough by itself to cause a release from adaptation in high-level visual processing regions. A size increase occurred once per block and never occurred on the first presentation. This task ensured that participants attended to the stimuli without requiring attention to object identity, keeping them visually engaged. Moreover, the size detection task encouraged participants to maintain central fixation throughout the scans as, regardless of size, the stimuli were always centered on the screen.

Preprocessing. All MRI data were processed using SPM12 and in-house matlab (Higham and Higham 2016) scripts. Functional data were first realigned and then slice-time corrected. Next, the functional images were co-registered to the structural image using a skull-stripped structural image as a source image. Both structural and functional images were then transformed into the standard Montreal Neurological Institute (MNI) space. No spatial smoothing was applied.

Functional Localizer and region-of-interest (ROI) identification. For each participant, a linear model was fit to the localizer run with a regressor for each domain (face, object, body part, scrambled images) as well as six movement parameters (x-translation, y-translation, z-translation, pitch, roll, yaw). The linear model was then convolved with the standard HRF function and a high-pass temporal filter of 128 seconds was applied. All functionally-defined regions were defined using

the Marsbar toolbox (Brett et al. 2002). We defined 18 functional ROIs (9 in each hemisphere), positioned on peaks of relative selectivity for objects, faces and body parts. A face > object contrast was used to define bilateral face-selective areas in the fusiform (posterior and middle fusiform face areas, FFA1 and FFA2, respectively (Pinsk et al. 2009; Weiner et al. 2014) and occipital gyri (occipital face area, OFA (Gauthier, Tarr, et al. 2000). An object > face contrast was used to define bilateral medial object-selective areas (MOA) in the parahippocampal and lingual gyri (Bi et al. 2016). By subtracting selectivity for faces, this contrast likely included regions that are often maximally responsive for scenes (which we did not use here) and least responsive for faces (Epstein et al. 1999). A body part > object contrast was used to define bilateral body part-selective areas in the fusiform (posterior and middle fusiform body area, FBA1 and FBA2, respectively (Peelen and Downing 2005) and occipital gyri (extrastriate body area, EBA, Downing et al. 2001). An object > scrambled contrast was used to define bilateral low-level object-selective regions in the lateral occipital complex (LOC, Grill-Spector et al. 1999b).

For each functional ROI, an in-house matlab script selected the peak functional voxel (27 mm³). Given previous research has found maximal reliability of individual differences in face selectivity at the functional peak of face-selective ROIs, without an advantage (and sometimes a decrease in reliability) of larger ROIs (McGugin and Gauthier 2016), this peak voxel was used for all subsequent analyses. See Supplementary Table S1 for average peak coordinates.

Adaptation effects. Adaptation effects were measured with a block design to ensure that the hemodynamic response for each domain was as strong as possible with as little noise as possible (Grill-Spector et al. 2006). For all ROIs, the raw signal for each adaptation run was extracted. For every data point, a pre-period signal was calculated by averaging the signal at the timepoints two seconds before the onset of every block and the timepoints at the onset of every block (Figure 2). This pre-period signal was calculated separately for alternating and repeating conditions and for each run. Every data point was then normalized using the respective average pre-period signal, with fMRI-adaptation quantified as percent signal change ($(\text{signal} - \text{pre-period})/\text{pre-period} \times 100$).

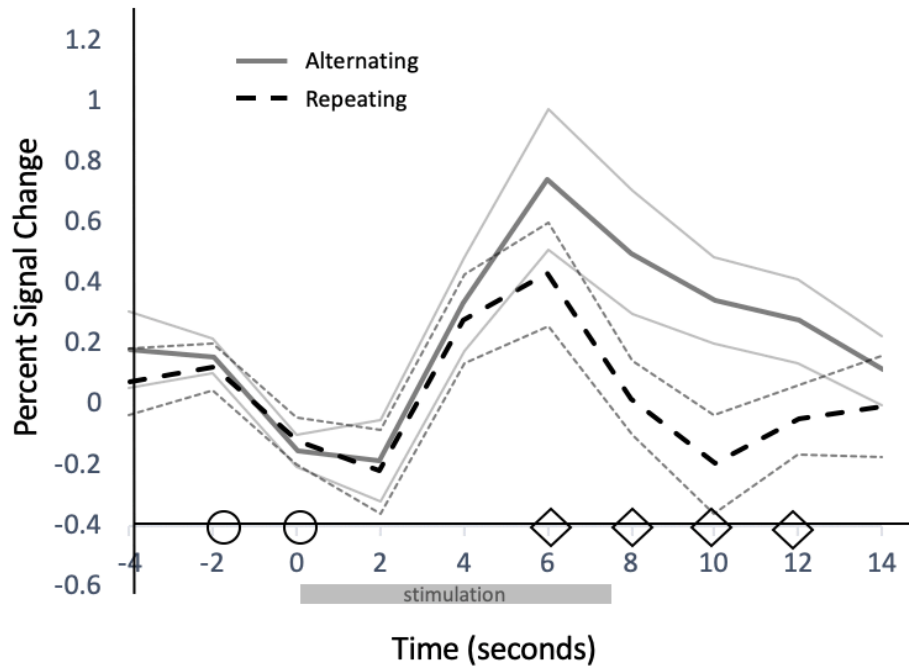


Fig. 2 | Percent signal change (PSC) averaged across runs and participants in the left posterior medial object area (IMOA1), where the largest adaptation effects were observed. The PSC time course is shown from 4 seconds prior to the start of a repeated or alternating block through 6 seconds after the end of the visual stimulation. Circles identify timepoints averaged for the pre-stimulus presentation PSC. Diamonds identify timepoints averaged for mean block percent signal change (PSC). 95% confidence interval depicted.

Bayesian Analyses. We use a Bayesian framework to quantify the relative support for different hypotheses. A Bayesian framework encourages the specification of competing models. Directional Bayesian tests do not suffer from the same limitations as directional frequentist tests, where no conclusion can be made when the evidence is in the non-predicted direction. In contrast, a directional Bayesian test is more sensitive to favorable evidence, but importantly, it is even more sensitive to unfavorable evidence (Wagenmakers et al. 2016). Bayes factors are best interpreted on a scale rather than through a cut-off, but to facilitate interpretation, we use Jeffreys’s conventions (Jeffreys 1998), characterizing BFs between 3 and 10 of substantial evidence for H_1 relative to H_0 , BFs between 10 and 30 of strong evidence, BFs between 30 and 100 of very strong evidence and BFs above 100 as decisive evidence. We used JASP (JASP 2020) to conduct these analyses. For correlation tests, we assigned an uninformative stretched beta prior (Ly et al. 2016) with a width of 1. For t-tests, we assigned δ a Cauchy distribution prior centered on zero with a scale parameter of 0.707. Because the

choice of priors can influence the results of Bayesian analyses (Depaoli et al. 2020), we report the sensitivity of results to other reasonable priors (changing priors after prior sensitivity analysis is considered bad practice, but reporting sensitivity to priors is recommended).

Results

fMRI-Adaptation in visual areas. We measured neural responses during *repeating* blocks, where 8 images showing the same object were presented in sequence (although object and background varied at the pixel level), to neural responses during *alternating* blocks, where images of two different objects alternated for the same duration (8 sec). To maximize construct coverage, we used 26 pairs from 26 different object categories. Participants performed the same incidental task (size judgment, see Figure 1D) during both conditions. In many visual areas, the neural response to a specific image is reduced with repetition, and a region that is sensitive to a given property will show release from adaptation when this property is modified (Grill-Spector and Malach 2001; Kourtzi and Kanwisher 2001). Average adaptation effects were quantified as the difference in percent signal change (PSC) observed for alternating vs. repeating blocks (See Figure 2). As expected, adaptation was observed in several ROIs with substantial evidence in favor of a reduction of activity for the repeating condition relative to the alternating condition in 9 ROIs (Table 1). These effects are relatively robust to the choice of prior. We used a Cauchy prior of .707, which implies an expectation of a strong effect. Using a Cauchy prior of .5, favoring smaller effects, led to only one additional ROI (IFBA1) showing moderate evidence for adaptation ($BF_{+0} = 3.12$). Using a Cauchy prior of 1, favoring even larger effects, led to only one ROI (rFFA2) going from indecisive to showing moderate evidence against adaptation ($BF_{+0} = .29$). For readers less accustomed with Bayesian inferences, we also report that support for adaptation survived Bonferroni correction for a frequentist t-test in seven of the 9 ROIs (see Table 1).

Table 1

Adaptation, indexed by the difference in PSC for the alternating vs. repeated condition

ROI	Mean diff	95% C.I.	SD	Cohen's d	BF ₁₀	
IEBA	.10	[-.10; .30]	.58	.17	0.50	
IFBA1	.25	[.01; .49]	.70	.36	2.61	
IFBA2	.20	[.05; .36]	.45	.46	7.85	**
IOFA	.23	[.04; .43]	.57	.41	4.44	*
IFFA1	.24	[-.01; .50]	.74	.33	1.91	
IFFA2	.09	[-.02; .21]	.33	.29	1.27	
ILOC	.39	[.15; .64]	.71	.55	27.80	**
IMOA1	.41	[.23; .58]	.52	.78	884.46	**
IMOA2	.25	[.09; .43]	.47	.54	23.46	**
rEBA	.18	[.00; .35]	.51	.34	2.21	
rFBA1	.15	[-.03; .33]	.52	.29	1.25	
rFBA2	.15	[.00; .31]	.45	.34	2.18	
rOFA	.20	[.04; .36]	.47	.42	5.18	*
rFFA1	.23	[.07; .39]	.45	.51	15.45	**
rFFA2	.10	[-.14; .33]	.69	.14	0.39	
rLOC	.14	[-.07; .36]	.62	.23	0.78	
rMOA1	.25	[.09; .40]	.45	.55	26.52	**
rMOA2	.28	[.11; .44]	.49	.57	35.10	**

Note. For all tests, H1 specifies that the correlation is positive. The 95% CI is the bayesian credible interval. * indicates effects with at least moderate evidence favoring adaptation ** indicates effects that would in addition survive a Bonferonni correction for a directional frequentist test, at $p=.05$.

Behavioral performance in the size judgment and α . We tested for evidence of a correlation between behavioral performance in the size judgment and α during fMRI-A runs and found no conclusive support for or against this relationship ($r = .32$, 95% C.I. [-.01;0.58], $BF_{10}=1.17$).

Neural correlates of α in functional ROIs. We examined whether the neural response to object shape in the alternating condition, controlling for the visual response during the repeating condition, was related to α (see Table 2, Figure 3). We report Bayesian support for the partial correlation between α and the response in the alternating condition, regressing out the response in the repeated condition. In addition, we verified that each ROI in which support was found for a partial correlation

also demonstrated a moderate zero-order correlation between the alternating condition and ϕ in the same direction. Indeed, all ROIs with support for a partial correlation showed a zero-order correlation above .32. As such, we exclude the possibility of strong suppressor effects (Lynam et al. 2006), where a partial correlation with the alternating condition would emerge only as a result of the introduction of the repeated response as a covariate. In relating ϕ to the magnitude of release from adaptation to shape differences, we used regression rather than subtraction because this allows for the effect of any variability in the response to the adapted condition to be removed. Not only are subtraction scores notoriously less reliable (Peter et al. 1993) but variance in the control condition is allowed to contribute to the variance of difference scores, instead of being removed (Degutis et al. 2013). A regression approach finds ROIs where ϕ predicts neural responses in the alternating condition that cannot be accounted for by variability in the repeated condition.

Table 2

Pearson partial correlations between *o* and the PSC in the alternating condition, controlling for the repeated condition.

ROI	<i>r</i>	BF _{+o}		95% C.I.
IEBA	.40	6.63	*	[.09;.63]
IFBA1	.17	0.54		[.01;.47]
IFBA2	.15	0.47		[.01;.46]
IOFA	.28	1.48		[.03;.55]
IFFA1	.16	0.52		[.01;.47]
IFFA2	.10	0.36		[.01;.43]
ILOC	.49	30.74	**	[.18;.69]
IMOA1	.06	0.28		[.01;.40]
IMOA2	.37	4.14	*	[.07;.61]
rEBA	.49	33.01	**	[.18;.70]
rFBA1	.07	0.30		[.01;.41]
rFBA2	.17	0.57		[.01;.47]
rOFA	.47	20.30	**	[.15;.68]
rFFA1	.38	4.46	*	[.07;.61]
rFFA2	.24	0.92		[.02;.52]
rLOC	.45	13.69	**	[.13;.66]
rMOA1	.45	14.99	**	[.14;.67]
rMOA2	.38	4.67	*	[.07;.62]

Note. For all tests, H1 specifies that the correlation is positive. The 95% CI is the bayesian credible interval.

* indicate BF_s with at least moderate evidence

favoring a correlation. ** indicates effects that

would in addition survive a Bonferonni correction for a directional frequentist test, at $p=.05$.

Evidence for a relationship between *o* and neural sensitivity to visual shape (see Table 2 and Figure 3) was observed in 9 ROIs: 6 right hemisphere ROIs (rEBA, rOFA, rFFA1, rLOC, rMOA1 and rMOA2) and 3 left hemisphere ROIs (IEBA, ILOC and IMOA2). Bayesian support for these effects was robust to a change in prior (we used a non-informative default prior of 1, which does not favor any correlation, no qualitative change in the support was found with a prior of .5, which favors more moderate correlations). The correlation in 5 of the 9 ROIs would survive a Bonferonni correction for a frequentist test (see Table 2). The effects were clearly driven by responses in the

alternating condition: when we looked at the correlations of the neural responses in the repeated condition with o , we found no support for a positive correlation in any of the ROIs and support for a null correlation in 13 of the 18 ROIs). Some ROIs where neural selectivity to shape was related to o did not show an average fMRI-A effect (e.g., IEBA). Indeed, across all ROIs we find support for a null correlation between the effect-size of fMRI-A and the correlation with o ($r = -.03$, $BF_{10} = .31$). In other words, different factors influence how much more one area is sensitive to shape than another versus the relative sensitivity to shape across individuals.

Are the neural correlates of o in different ROIs related to a common mechanism? Conceptually, asking whether the correlation with o across ROIs survives a correction for multiple comparisons begs an important question: to what extent do these correlations of fMRI-A for novel objects with o in different ROIs reflect a common mechanism? Inspection of the correlations between residual neural responses (responses to the alternating condition, controlling for the repeating condition) across ROIs reveals many correlations above .3 (Supplementary Table S2). Correlations among residual neural responses in only the 9 ROIs where we found support for a correlation between those responses and o were equal or larger than .3 in 75% of the cases (and averaged to $r = .39$ after Fisher Transform). The residual neural responses in all 18 ROIs were submitted to a Principal Components Analysis (PCA). The PCA provided evidence for a first factor accounting for 35.2% of the variance, with a Kaiser-Meyer-Olkin (KMO) measure of sampling adequacy of .52. Six factors in total had eigenvalues above 1 (5.77, 2.06, 1.90, 1.53, 1.12 and 1.03) and together accounted for 74.5% of the variance (35.2%, 11.5%, 10.5%, 8.4%, 6.2% and 5.7% respectively).

We found decisive evidence supporting a correlation between o and the 1st factor of the PCA on neural effects ($r = .55$, 95% C.I. [.25;.73], $BF_{+0}=121$; Figure 3). In contrast, we found no support for any of the other factors to be correlated with o ($r_s < 0$, all $BF_{+0} < .4$). While the neural correlates of o are distributed among visual areas, they can be conceived as largely reflecting a common

mechanism.

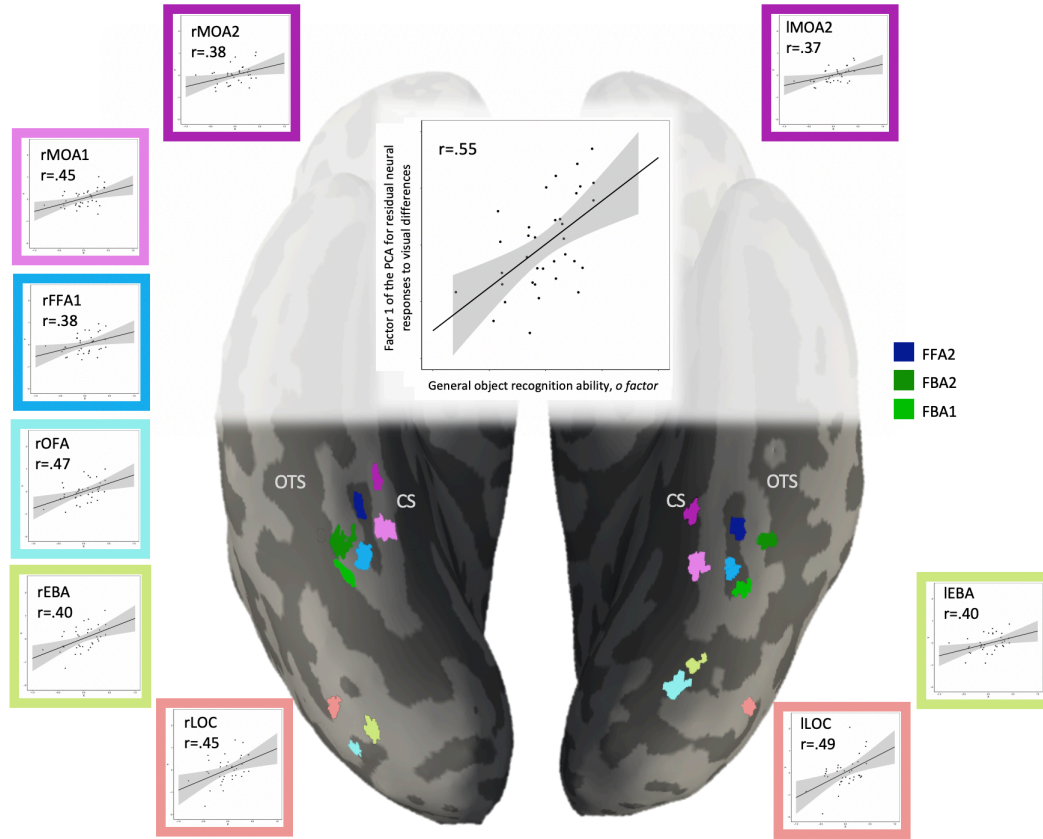


Fig. 3 | Central panel: Correlation between the first factor of the PCA for residual neural responses to visual differences and *o* factor scores. The shaded region shows the 95% confidence interval. Surrounding panels: scatterplots for individual ROIs with at least substantial evidence in favor of a positive partial correlation between residual neural responses to visual differences with *o* factor scores. Three ROIs (FFA2, FBA1 and FBA2) showed no such evidence in either hemisphere. Per radiological convention, the RH is shown on the left. The two hemispheres come from different participants and are chosen to illustrate variability in the location of occipital ROIs. In particular, the OFA and EBA had the most variability in location across subjects. The OFA was generally posterior to the LOC ROI, as shown in the RH example (in 72% of subjects). In the 10/36 subjects with an OFA anterior to the LOC (as shown in the LH example), 9 of those also have an EBA anterior to the LOC. The majority of subjects had the same pattern in both hemispheres (LOC relative to OFA, 64%; LOC relative to EBA, 78%).

Whole brain search in gray matter for the correlates of *o*. To further explore whether functional correlates of *o* extend beyond functionally-defined regions of interest, we conducted a voxel-wise analysis, after imposing an explicit mask to include only gray matter (7223 voxels, 157.5 resels). We created an explicit grey matter mask to constrict analyses (excluding activations in the cerebellum, white matter, and ventricles) and thereby increase our sensitivity to cortical effects. However, no

significant clusters survived the lowest levels of correction for multiple comparisons. At a very liberal uncorrected threshold of $p < .01$ with a cluster threshold of only 3 voxels, only 5 clusters were found in parietal and frontal areas (while unlikely to replicate, the coordinates are provided in supplemental material).

How specific to category-selective areas is the correlation to o ? Aside from the fact that testing for effects in 18 category-selective areas required a less stringent level of correction for multiple comparisons, individually-defined functional ROIs could also be more sensitive if responses are small and variable in location across subjects, but better aligned across subjects within category-selective foci. We conducted two additional analyses to determine whether the neural correlates of o were especially centered on category-selective foci.

Individual differences associated with experience can be highly focused to category-specific areas. For instance, expertise effects for cars are highly specific to the center of the FFA, with a correlation between behavior and neural selectivity above $r=.6$ inside a 200 mm^2 area but falling sharply to $r \sim 0$ beyond this boundary (McGugin et al. 2012). To see if the present effects have such sharp boundaries, we compared the correlations with o in the original small ROIs of 3mm radius (approximately 113 mm^3) to those in larger hollow ROIs that surround these peaks (9mm – 6mm radius (approximately 2148 mm^3) and 15mm – 12mm radius (approximately 6896 mm^3) (see Figure 4; supplementary Table S3). We identified four distinct patterns when examining the correlates of o in large ROIs distant from peaks of category specificity: out of 9 ROIs for which we found support for a correlation in the center, 6 (rEBA, rOFA, rMOA1, rLOC, lEBA and lLOC) showed a very similar or larger correlation at a distance from the peak (Pattern A). The other 3 ROIs (rMOA2, lMOA2 and rFFA1) show a drop in correlation moving away from the peak (Pattern C). Of the 9 ROIs for which we found no clear support for a correlation at the peak, 6 (rFFA2, rFBA2, lFBA1, LOFA, LFFA1 and lMOA1) eventually showed a correlation in the largest ROI (Pattern B). Finally, there were 3 ROIs for which no correlation with o was observed at any size (Pattern D).

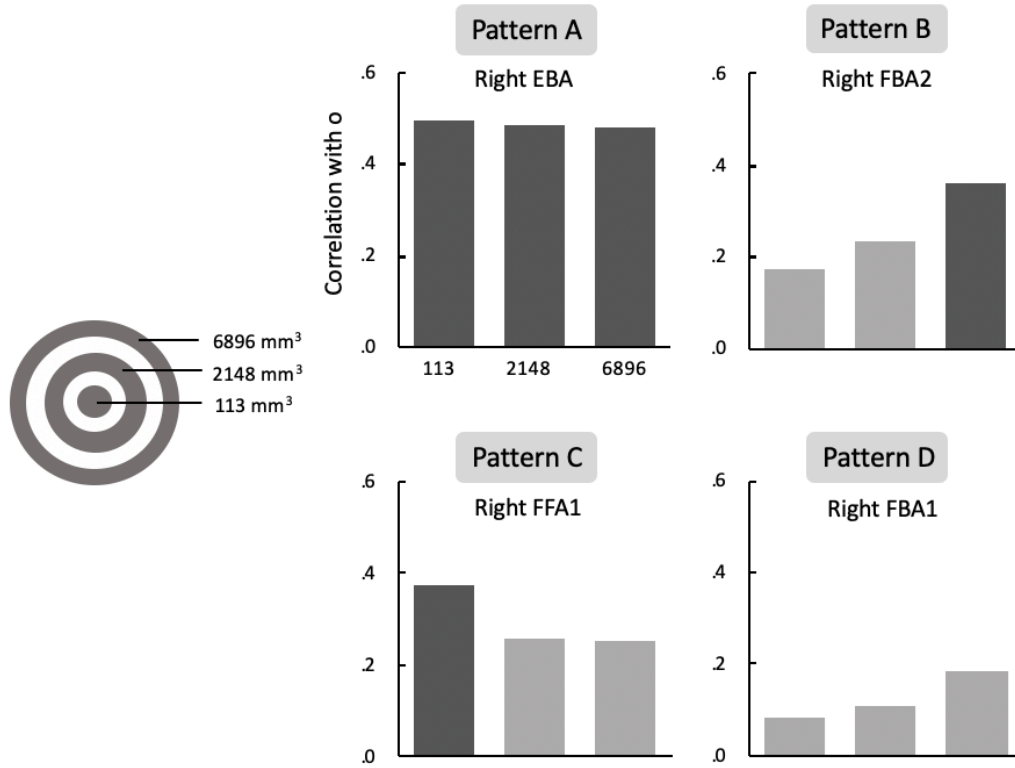


Fig. 4 | Results from increasing the distance from the ROI peak, shown in four patterns of change in correlation with o . Left. Schematic representation of ROIs centered on the peak of functional selectivity (3 mm radius, $\sim 113 \text{ mm}^3$), and expanding beyond into larger hollow ROIs (6 – 3 mm radius, 9 – 6 mm radius, 12 – 9 mm radius, and 15 – 12 mm radius). One representative ROI from each pattern is plotted. See supplementary Table 4 for all correlations. Dark bars indicate a significant correlation with o ($BF_{+0} > 3$).

Thus, unlike domain-specific effects of expertise (McGugin et al. 2012), neural correlates of o are mostly expressed in larger ROIs that are centered on these peaks. Next, we wanted to know if larger ROIs that were not defined individually on the basis of localizers would also capture the correlation between neural shape selectivity and o . To ask this question we analyzed correlations of neural selectivity with o in 66 cortical anatomical ROIs available in the SPM toolbox (Tzourio-Mazoyer et al. 2002). Due to the large multiple comparison problem with our relatively small sample, we do not report inferential statistics and only report effect sizes.

The pattern of correlations across 66 anatomical ROIs was quite systematic (and correlations were robust to potential outliers, as supported by skipped Pearson correlations (Wilcox 2004). Out of 13 ROIs showing effect sizes above $r = .3$ (Table 3), 10 were in occipitotemporal cortex (Figure 5). All of these ROIs also showed moderate zero-order correlations between o and the response in the alternating condition (ranging between $r = .27$ and $r = .45$). The list includes regions in early visual cortex (right cuneus and right lingual), even though these areas showed no sign of fMRI adaptation on average – as expected based on the fact that even repeated objects were slightly rotated on each presentation (Murray et al. 2006). In other words, and just as in some of the functional ROIs, variability in adaptation was in some cases correlated with o in areas showing no average fMRI-adaptation. This was also true in non-visual areas, such as the left precentral cortex and the right superior parietal cortex, areas that were also engaged by informative visual fragments in two studies (Lerner et al. 2008; Holzinger et al. 2019).

Table 3

Anatomical ROIs showing partial correlations between o and the PSC in the alternating condition, controlling for the repeated condition, and mean response and difference between conditions.

ROI	Corr with o	mean ALT	sd ALT	mean REP	sd REP	mean ALT-REP	sd ALT-REP
r middle occipital	.53	.28	.22	.13	.19	.15	.23
l middle occipital	.50	.20	.24	.08	.19	.12	.21
r fusiform	.45	.20	.19	.06	.12	.14	.21
r cuneus	.44	-.22	.17	-.09	.22	-.13	.24
l inf. occipital	.43	.48	.31	.22	.22	.26	.27
l sup. occipital	.42	.03	.21	.02	.19	.01	.23
r sup. occipital	.42	.09	.21	.04	.18	.05	.22
r inf. occipital	.41	.56	.36	.32	.23	.23	.32
l fusiform	.39	.17	.17	.02	.14	.15	.21
r sup. parietal	.32	.04	.20	.03	.17	.01	.28
l precentral	.32	.04	.13	.03	.11	.00	.18
r lingual	.32	-.05	.19	-.01	.18	-.04	.25
l sup. parietal	.30	.07	.18	-.01	.18	.09	.27

note. Only ROIs with a correlation of .3 and above (arbitrary threshold) are listed. Full list of 66 ROIs in supplementary Table 5, which also includes skipped correlations, to show robustness to possible outliers.

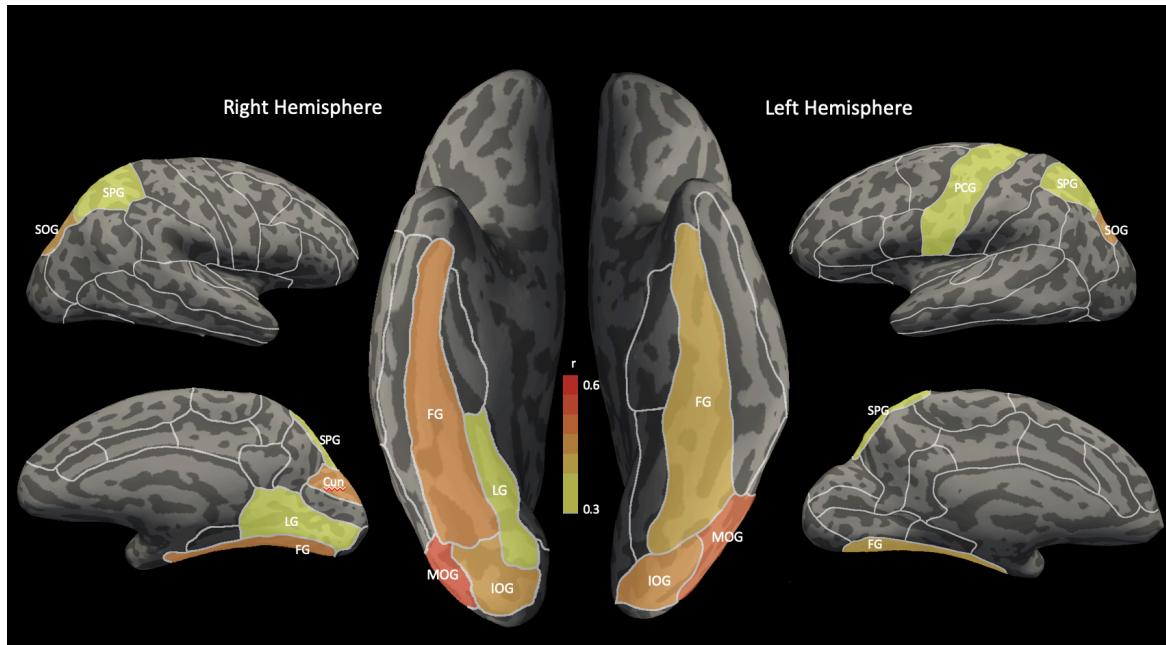


Fig. 5 | Anatomical ROI analysis. Anatomical ROIs are outlined in white across both hemispheres and shown in three views (center: inferior; top: lateral; bottom: medial). This is an exploratory analysis: the 13 ROIs with correlations with ρ greater than or equal to $r=.3$ are shaded with a color to indicate the strength of the correlation. Labeled ROIs: Fusiform Gyrus (FG), Lingual Gyrus (LG), Middle Occipital Gyrus (MOG), Inferior Occipital Gyrus (IOG), Cuneus (Cun), Superior Occipital Gyrus (SOG), Precentral Gyrus (PCG).

Discussion

We find that a domain-general ability related to the visual recognition of objects has neural correlates distributed across early visual cortex, ventral visual areas in occipitotemporal cortex, and parietal areas also implicated in shape perception. Individuals with a higher ρ displayed higher sensitivity to object shape in these areas. The relationship between ρ , which captures variance common across behavioral tasks and categories (both novel or familiar), and neural sensitivity to shape across several parts of occipitotemporal cortex cannot be explained by an overlap in task or category. Prior work relates performance on a visual task to the neural responses evoked by that task in the scanner (Grill-Spector et al. 1999b). Additional work links behavioral performance for a given category in separate behavioral tasks to neural selectivity for that same category in specific visual areas (Wong et al. 2009; Natu et al. 2016). Here, there was no overlap between stimuli or tasks used in the scanner and those tasks used to estimate ρ . Accordingly, we expect our effects to generalize to

a wide range of tasks and categories, providing a framework to predict visual learning based on variability in neural responses.

Neural correlates of o were identified in large visual ROIs but not in a voxel-based analysis of all gray matter, suggesting that patterns are not only distributed but vary across participants. However, the only analysis in which the effect survived correction for multiple comparisons is the analysis in functionally-defined category-selective ROIs. This is not unexpected: while we defined 18 functional ROIs, there were 66 anatomical ROIs and 157.5 resels in the voxel-based analysis. Recent work investigating the statistical power in fMRI studies of brain-behavior relationships, outside of an ROI approach, suggests that much larger sample sizes (>80) may be required to identify neural correlates of behavior and produce stable effect sizes (Grady et al. 2021). Our search for the neural correlates of o in category-selective areas was originally motivated by the fact that on the one hand, these include areas where correlations with behavioral performance with complex objects have been described before (e.g., Xu 2005; McGugin et al. 2012, 2014), and on the other hand, it was a successful approach in studies looking at regions engaged by the best features for object recognition (Lerner et al. 2008; Holzinger et al. 2019).

We found that no type of category-selective area (e.g., face vs. object selective) was more likely to show a correlation of neural selectivity with o , nor was the correlation with o in these regions highly focal. Yet, even if category-selective peaks were not uniquely associated with the effect, defining functional ROIs turned out to be an effective means of detecting it. That is, no cluster of activity correlating with o was found in the ventral pathway in the voxel-based analysis, even at an uncorrected $p < .01$ level. This is consistent with evidence that the advantages of functional ROIs go beyond statistical power, as they have been shown to improve sensitivity, functional resolution and provide more accurate effect sizes (Nieto-Castañón and Fedorenko 2012). Future work may compare the measurement of the neural correlates of o using different methods to align brains which vary in anatomy and functional organization both across and within cytoarchitectonic areas. Such methods include cortical folding-based alignment (Frost and Goebel 2012), multimodal surface matching which uses a combination of anatomical and functional features as well as connectivity

(Robinson et al. 2014), and hyperalignment based on functional responses during movie viewing (Haxby et al. 2011).

While better inter-subject alignment could improve sensitivity of voxel-based analyses, one aspect of our results suggests that determining the specific voxels in visual areas in which activity is related to o may not be the most meaningful question, because we found that a common source of variance underlies the correlation across all the ROIs in which it was found. This is significant given recent claims that there may be no objective way to establish one-to-one correspondence for different clusters, even within the same domain of face-selectivity (Gao et al. 2022). In the present case, because the effect of interest is the same across clusters, a mis-correspondence would be less problematic. Thus, because this is not an effect that by definition can be localized in a single individual and because it is highly distributed, various applications may seek to simply measure this relationship without seeking precision regarding its spatial extent. For instance, a study may benefit from quantifying domain-general effects in a region where a training effect is predicted, to investigate the relationship between individual differences before and after training. Or a researcher may postulate that the correlation between neural shape selectivity and o should be modulated by the task performed in the scanner. Such research questions would benefit from using category-selective ROIs in the same way that the ROI approach has facilitated other research endeavors (Nieto-Castañón and Fedorenko 2012). Another approach to increase statistical power would be to use a subset of the anatomical ROIs that produced the largest effect sizes in this study. This approach would be supported by the similarity between the set of areas showing the largest correlations and those recruited by the “best” features for shape processing in prior work (Lerner et al. 2008; Holzinger et al. 2019).

These fMRI studies reporting on the response evoked by the most informative visual features included distributed activity in ventral occipitotemporal cortex, the superior parietal gyrus, and premotor cortex. In prior work, authors speculated about the reasons for the best features to engage parietal (attentional or semantic effect) and prefrontal (action planning) regions. Our use of novel objects would limit the role of semantic knowledge, which may have been engaged in studies with familiar categories. It seems reasonable that people with high o could be particularly efficient at

finding and encoding the most informative features when viewing objects, such that individual differences in o are related to areas that on average represent the features most useful for object recognition. However, the similarity of the pattern of o -related areas and that for informative features is at the moment only qualitative and should be investigated within the same study.

In contrast, the correlation of o with neural sensitivity to shape was not related to the relative magnitude of adaptation responses to shape across areas. It may not be surprising that different factors determine the mean responsivity to shape across different visual areas and individual differences in sensitivity to shape within each area. Indeed, the range of stimuli we used to measure sensitivity to neural shape was large, such that we would expect similar individual differences to replicate with a different set of objects. In contrast, the degree of adaptation in different areas is highly dependent on the stimulus set (e.g., Gauthier, Tarr, et al. 2000; Chouinard and Goodale 2012).

Our irrelevant task ensured equal task-driven attentional requirements in the alternating as the repeating condition. But it is possible that individuals with high o , by virtue of being more sensitive to differences in shape, could have experienced more bottom-up attentional capture. In that sense, visual ability may be difficult to dissociate from its downstream effects. Another line of work that converges with ours concerns haptic shape processing. Recent work found that o measured in visual tasks correlates with haptic object recognition performance (Chow et al. 2021), suggesting the ability could be amodal. Haptic shape perception recruits ventral and dorsal visual high-level areas (Snow et al. 2014; Lee Masson et al. 2016). Early visual areas have been tied to the processing of fine shape information in both visual (e.g., Smith & Goodale, 2015) and haptic tasks (e.g., Lee Masson et al., 2016). For the first time, our work illustrates that those with the best shape processing ability (possibly an amodal ability) engage a distributed network of shape-sensitive areas more than those with poor abilities, during an incidental visual shape processing task.

Our work illustrates important differences between individual differences and experimental effects that have been discussed in the behavioral literature (e.g., Goodhew & Edwards, 2019; Hedge et al., 2018). Brain areas, either small functional ROIs or larger anatomical ROIs, where we found the neural correlates of o overlap to some extent with those that show average fMRI-A to shape, but they are not identical. Many research programs begin by the study of average effects, and in the case

of brain imaging, identifying brain areas for these average effects. But individual differences need not be limited to the areas that are recruited on average, as individuals with higher abilities could have more distributed visual responses (Miller and Schwarz 2018).

The incidental nature of our measurement of neural selectivity to shape may have led to particularly distributed neural correlates of *o*. Goals influence object processing and its neural substrates (e.g., Harel et al., 2014), and the neural correlates of *o* may likewise be influenced by the task used in the scanner. One factor that can influence the distributed nature of the neural correlates of behavioral performance is the difficulty of the fMRI task. For instance, under low perceptual and attentional load, correlates of car expertise were found in several areas, but when measured in a task with higher load, the only robust correlation with expertise was found in the right FFA2 (McGugin et al. 2014). Thus, it is possible that if neural effects were measured in a task requiring explicit attention to shape, the pattern of correlations with *o* could also be more limited.

Extrapolating from our results to the processing of familiar objects, it is critical to take variability in experience into account when predicting neural correlates of performance. The neural correlates of *o* should be relevant across a wide diversity of conditions at low levels of experience, but the pattern of selectivity for a category-specific ability may narrow as experience grows. Using very similar behavioral tasks and an fMRI adaptation paradigm with faces, developmental work found that neural sensitivity to the shape of faces increased with age in face-selective, but not object selective, areas (Natu et al. 2016). This neural sensitivity to faces in face-selective areas, across both children and adult participants, was associated with perceptual discrimination for faces. Neural mechanisms that have been invoked to account for changes with age and experience include narrowing of the tuning of neuronal responses (Freedman et al. 2006) or increases in the number of visual neurons responding to a category (Kobatake et al. 1998). Early experience with faces appears to be required for the development of face patches in monkey cortex (Arcaro et al. 2017) and training with non-face objects increases both neural selectivity in visual areas and perceptual discrimination (Gauthier et al. 1999; Wong et al. 2009; Folstein et al. 2013). Specialization due to experience may operate on responses that are to some extent driven by physical characteristics, but these dimensions do not fully determine selectivity to categories. Differences in training regimen

with the same objects result in qualitatively different patterns of specialization (Wong et al. 2009, 2012) while the same area can show selectivity for objects with widely different geometry, based on similar experience (Xu 2005; McGugin et al. 2018; Burns et al. 2019). Individual differences that relate to o may represent the springboard from which these effects can grow.

This prediction stems from an analogy to Cattell's Investment Theory (Cattell 1971) suggesting a role of fluid intelligence in the acquisition of crystallized intelligence and its later modification that involved learning as a mediator of this relationship (Schweizer and Koch 2002). A high-level of fluid intelligence tends to predict not only differences at pre-test, but also the rate of learning from experience (Kvist and Gustafsson 2008; Primi et al. 2010). Fluid intelligence, not unlike o in our results, has a distributed neural substrate. Interestingly, the brain volume of several brain regions associated with fluid intelligence (e.g., middle frontal, anterior cingulate, caudate) predicts not only pre-intervention differences but also the magnitude of the change due to a range of interventions (Daugherty et al. 2020). Here we speculate that o may influence category-specific abilities and the category-specific neural specialization, mediated by learning. In such a framework, properties of the stimuli and experience may dictate *where* specialization happens for a given task, but o could predict the *magnitude* of the neural changes in learning-relevant regions, as a result of this experience.

This framework can also help explain how variability in domain-general performance can result in category-specific deficits. Recent work using a fast periodic visual stimulation EEG paradigm found reduced neural sensitivity to the identity of both faces and cars in individuals with congenital prosopagnosia (Fisher et al. 2020). This was a puzzling result because the behavioral deficit compared to controls was relatively specific to faces. Such a pattern could emerge if the visual responses evoked when seeing objects reflect the distributed response related to o that we see here. If individuals with congenital prosopagnosia have a low o , they could benefit much less than controls from experience with faces (whereas both groups would have limited experience individuating cars). Accordingly, a low o in congenital prosopagnosia could result in a reduced distribution of response to any category (Jiahui et al. 2018). This would explain why the magnitude of difficulties with faces and objects is correlated in this population (Geskin and Behrmann 2018).

But a low o during development would limit learning from experience with faces, and because face recognition experience is relatively universal, this then results in a category-specific deficit.

The importance of our results is rooted in the psychometric and modeling work offering strong evidence for a domain-general object recognition ability (Richler et al. 2019; Sunday et al. 2021). Other efforts have found evidence of general factors in visual processing. One study found a general learning ability that accounted for about 37% of the variance among 39 subjects, across a wide range of perceptual learning tasks (Yang et al. 2020). Another study on more than 36,000 individuals playing various games on the Lumosity cognitive online training platform found evidence of a general ability factor that reflected mostly later stages of practice (Steyvers and Schafer 2020). Both general factors, however, appeared related to general intelligence, or g . In contrast, while g can account for a small amount of variance in individual tasks used to estimate o , the majority of the shared variance across object recognition tasks is independent of g (Richler et al. 2017, 2019). That o can account for individual differences independent of g suggests the potential to add predictive value to what is achieved with already powerful dimensions of human behavior, such as intelligence and personality. For instance, o accounted for an additional 15% of the variance in learning to detect lung nodules in chest radiographs, after controlling for both experience in this domain and g (Sunday et al. 2018).

We acknowledge the limitations of our work, aside from its sample size. On the behavioral side, a latent factor like o can only be interpreted within the context of the specific indicators included in its estimation. More specialized factors for memory vs. perception tasks, suggested in work modeling face recognition abilities (Hildebrandt et al. 2010; Wilhelm et al. 2010) could isolate sub-networks of brain areas. On the neural side, while we can draw inferences from the correlations that we do observe, we should be very careful not to draw strong inferences from null effects in other regions. For instance, it is possible that other functional localizers, even in non-visual areas, could reveal other neural correlates of this domain-general ability. Finally, while we made several design choices based on best practices for individual differences measurement (Richler et al. 2014; Hedge et al. 2018; Goodhew and Edwards 2019), we did not compare these choices to more standard approaches and therefore cannot quantify the advantages they may have afforded in an fMRI setting.

Future work providing such comparisons could help meet well-known challenges in the use of fMRI measures for individual differences research (Elliott et al. 2020).

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